FULL STACK / REPORTING & DATA / AI-ML

Ushan Jayamanne

Senior Software Engineer · Omobio (Pvt) Ltd

Reliable systems for millions of messages: telecom platforms, real-time reporting & data, and AI/ML.

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3+	40K	6	12
Years of experience	Peak TPS handled	Enterprise clients	Industry projects delivered

ABOUT ME

An engineer who creates value

Senior Software Engineer with 3+ years building enterprise telecom messaging platforms, real-time reporting and data pipelines, and full stack web applications. I lead the engineering team on the Banglalink SMSC platform, owning architecture, performance and production stability. I combine Python, PHP, ReactJS and Node.js with ClickHouse data engineering and machine learning for SMS firewall campaign detection, and I've delivered for Banglalink, Mobitel, Airtel, Ncell, Hutch and Hemas Hospital.

EXPERIENCE

Trainee to team lead in three years

Jun 2026 – Present
Senior Software Engineer
Omobio (Pvt) Ltd

- **Team leadership:** lead the engineering team on the Banglalink SMSC / CBVC platform through architecture design, sprint delivery and code review.
- **Billing engine:** designed ClickHouse event tables with minute → year materialized views and a reusable Python report engine serving 70 reports (JSON API for PHP).
- **AI/ML firewall:** built the real-time campaign detection pipeline (LightGBM, HDBSCAN, Sentence Transformers, LLM verification) with Docker and Jenkins CI/CD.
- **Performance:** cut Vector + ClickHouse CDR ingestion CPU from ~90% to target and met the 4-minute write SLA under load.

Mar 2024 – May 2026
Software Engineer
Omobio (Pvt) Ltd

- **Banglalink SMSC:** front-end and back-end features across 16 servers at up to 40,000 TPS, plus automated Python traffic reporting.
- **Real-time analytics:** Vector → ClickHouse → Apache Superset pipeline with KPI dashboards for success rate, errors, TPS and operator traffic.
- **Healthcare:** Hemas Hospital Admin Portal in ReactJS and payment-failure reporting in Directus Flows.
- **Change requests:** Mobitel SMSC, Hutch HOPP and Airtel Self Care App.

Sep 2023 – Mar 2024
Associate Software Engineer
Omobio (Pvt) Ltd

- **Airtel Jersey SMSC:** full stack development (Ext JS / PHP) covering new features, change requests and stability.
- **OmobioNG website:** built the company's public website in SvelteKit.

Mar 2023 – Sep 2023
Trainee Full Stack Developer
Omobio (Pvt) Ltd

- **Ncell & Airtel Jersey:** bug fixes, change requests and integration testing on live SMSC systems using ReactJS, PHP, MySQL and Oracle DB.

Turning heavy data into real outcomes

Production systems for telecom operators and healthcare, newest first.

SMS Messaging Hub · Apr 2026 – Present

Reporting & Billing Engine

The backend that turns huge volumes of message events into fast reports and accurate billing data. Role: designer and lead developer.

events → min → hour → day → month → year

- **Data model:** ClickHouse tables for message, audit and SMPP data; heavy maths runs once per minute and each level sums the one below.
- **Report engine:** each report is just SQL; JSON in/out for PHP, pagination and totals in one query, injection-safe parameter binding.
- **Catalogue:** 70 reports across administration, alerts, finance, rates and reference data.
- **Quality:** Docker with tracked auto-migrations, Conventional Commits and SonarQube.

ClickHouse · Python · PHP · Docker · SonarQube

AI / ML — Apr 2026 – Present

Real-time AI/ML Campaign Detection

Catches spam and malicious SMS campaigns from 5-minute CDR batches.

- Sentence Transformers + HDBSCAN cluster look-alike messages.
- LightGBM scores campaigns in under 1 ms across three risk levels.
- LLM verification; Docker and Jenkins CI/CD.

LightGBM · HDBSCAN · Python · Jenkins

REPORTING & DATA — Jan – Apr 2026

Banglalink SMSC – Dynamic Reporting

Real-time CDR analytics for SMSC and Bulk SMSC.

- Vector → ClickHouse → materialized-view aggregators.
- Superset dashboards: success rate, errors, TPS, operators.
- Tuned from ~90% CPU to target; 4-minute write SLA met.

ClickHouse · Superset · Vector · Linux

TELECOM — Mar 2025 – May 2026

Banglalink SMSC – Product & Reports

Core SMS centre for one of Bangladesh's leading operators.

- Features across 16 servers, incl. A2P roaming DB migration.
- Automated Python reporting at up to 40,000 TPS.
- MariaDB SQL for partitioned tables and TPS KPIs.

Ext JS · PHP · Python · MariaDB

WEB APP — Feb – Oct 2024

Hemas Hospital Admin Portal

One portal for hospital staff to run digital services.

- App controllers, home care, virtual care, doctors, customers.
- Payment-failure reporting flow in Directus.

ReactJS · REST APIs · Directus

TELECOM — Jan – Mar 2025

Mobitel SMSC

Front-end (Ext JS) and back-end (PHP) developer on the SMSC product across 2 servers: traffic handling, reports, features and maintenance.

Ext JS · PHP · SQLite

TELECOM & WEB — 2023 – 2026

Other Client & Company Projects

- **Airtel Jersey SMSC** – features, CRs, stability.
- **Hutch HOPP & Airtel Self Care** – PHP CRs.
- **Ncell** – ReactJS / PHP features and fixes.
- **OmobioNG** – company website in SvelteKit.

Ext JS · PHP · ReactJS · SvelteKit

PERSONAL

Full Stack Side Projects

2022 – 2023

YouTube-style video app with uploads to Firebase Storage, subscriptions, comments and search; MERN **social media** and **blog** apps with REST APIs and auth; Vue.js restaurant and e-commerce front ends; JavaFX student management system.

MongoDB · Express.js · React · Node.js · Firebase · Vue.js · JavaFX

Everything I work with

Core skill

</> Languages

Python · PHP · JavaScript · SQL · TypeScript · Java · HTML5 / CSS3

UI Front-end

ReactJS · Ext JS · SvelteKit · Next.js · Vue.js · Responsive UI

API Back-end & APIs

Python · PHP MVC · REST APIs · Node.js · Express.js · FastAPI · Microservices

DE Reporting & Data

ClickHouse · Materialized views · Apache Superset · Vector · ETL · Data modelling · CDR analytics

DB Databases

ClickHouse · MariaDB · MySQL · MongoDB · Oracle DB · SQLite · Redis

ML AI / ML

LightGBM · HDBSCAN · Sentence Transformers · XGBoost · scikit-learn · LLMs

Ops DevOps & quality

Docker · Git · Linux · Jenkins CI/CD · GitHub Actions · SonarQube · pytest

SMS Telecom

SMSC · SMPP · CDR processing · A2P / Bulk SMS · SMS firewall · Billing

AI Tools & ways of working

Team leadership · Agile / Scrum · Code review · Claude AI · ChatGPT · Gemini · Jira

Education

2019 – 2023

BSc in Computer Science

Eastern University, Sri Lanka – Trincomalee Campus

2019

BIT – First Year Diploma

ESOFT Metro Campus

2019

Aquinas Diploma in English

Joseph Vaz Institute, Kurunegala

2017

GCE Advanced Level – Maths

Combined Maths A · Physics B · Chemistry S

REFERENCE

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LET'S BUILD TOGETHER

See all 18 projects on my portfolio

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